Campus Crowd Predictor

BYOP Capstone Project Report

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1. Introduction

One problem I noticed as a first-year student is that it is difficult to know in advance how crowded a campus facility — like the gym, library, or cafeteria — will be at a particular time. Most students end up going and finding the place packed, which wastes time. I thought this was a practical problem I could solve using Python and some basic data analysis concepts covered in our course.

The idea behind the Campus Crowd Predictor is simple: if we collect historical data on how many people visit a facility at different times and days, we can use that data to give a reliable estimate of how busy it will be in the future. This is a real-world application of data-driven prediction, which is one of the core ideas in machine learning.

2. Approach and Methodology

Since I did not have access to real footfall data from my campus, I wrote a data generation script in Python that simulates realistic crowd patterns based on common observation. The generated dataset covers 90 days and includes three facilities: Gym, Library, and Cafeteria. Each row in the dataset represents one hour at one facility on one day, with a headcount and a crowd status label.

The crowd patterns were designed to reflect real behaviour [1]:

- The gym is most crowded in the evening (5 PM to 8 PM), which matches typical student schedules after classes.
- The library peaks during mid-morning to afternoon (10 AM to 3 PM), when students are studying between lectures.
- Weekends see significantly lower footfall across all facilities.

The predictor script reads this CSV dataset and, for a given facility, day, and time, averages all historical headcounts for that exact combination. It then classifies the result as Empty (under 25 people), Moderate (25 to 59), or Crowded (60 or more). This is a simple but effective form of instance-based prediction, similar in spirit to a k-Nearest Neighbour lookup [2].

3. Key Decisions Made

A few decisions shaped how the project came together:

- I chose to generate synthetic data rather than skip data collection entirely, because having a realistic dataset makes the predictor genuinely useful and testable.
- I kept the project to pure Python with no external libraries. This made it easier to understand each part of the code and kept the setup simple for anyone running it.
- I used a dictionary-based data structure to index records by facility, day, and time. This made lookups fast and the code easy to read.
- I added operating hours logic so the tool returns a meaningful CLOSED message instead of failing silently for invalid time queries.

4. Challenges Faced

The biggest challenge was making the synthetic data feel believable. In real life, crowd patterns are affected by exams, holidays, weather, and events — none of which I could easily model. I kept the simulation straightforward and focused on the most consistent patterns (time of day, day of week) rather than trying to over-engineer it.

Another challenge was handling edge cases in the predictor, such as times that fall outside operating hours or queries with slightly different formatting. I added basic input handling to catch these gracefully.

5. What I Learned

This project taught me how much of data science is just careful data organisation. Before you can make any prediction, you need clean, well-structured data — and building the generator helped me understand what that actually means in practice.

I also got a much clearer understanding of how simple averaging can serve as a baseline prediction model. It is not machine learning in the complex sense, but it is exactly the kind of pattern-matching that ML builds on [3]. Starting here made the bigger concepts feel more grounded.

Finally, working through the full cycle — problem, data, code, output, documentation — gave me a sense of what a real software project looks like beyond just writing functions for assignments.

6. Conclusion

The Campus Crowd Predictor is a small but complete project that addresses a genuine everyday problem using Python fundamentals and basic data analysis. It demonstrates data generation, file I/O, dictionary-based data structures, and a simple prediction mechanism — all concepts from our course. With real data from a campus management system, this tool could be made practically useful with minimal changes.

6. screenshots:

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PS D:\byop_folder> & C:/Users/admin/AppData/Local/Programs/Python/Python313/python.exe
 d:/byop_folder/predictor.py
--- Campus Crowd Predictor ---
Enter facility (Gym, Library, Cafeteria): gym
Enter day (e.g. Monday): tuesday
Enter time (e.g. 14:00): 13:00

Calculating...
Status: Empty
Average people: 20.9
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PS D:\byop_folder>
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PS D:\byop_folder> & C:/Users/admin/AppData/Local/Programs/Python/Python313/python.exe
 d:/byop_folder/predictor.py
--- Campus Crowd Predictor ---
Enter facility (Gym, Library, Cafeteria): cafeteria
Enter day (e.g. Monday): sunday
Enter time (e.g. 14:00): 02:00

Calculating...
Status: CLOSED (Cafeteria hours are 10:00 to 01:00)
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PS D:\byop_folder>
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